

Evaluating AI-Assisted Decision Augmentation Systems

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CSCI 5310: Decision Support Systems

Research Project: Evaluating AI-Assisted Decision Augmentation Systems

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Task 1: ED² Statin Choice – Medical Decision-Making and Reactive Problem-Solving

1. Brief Description of Case 1

Marcus is a hypothetical 48-year-old engineering manager who was just flagged at his annual physical for elevated cholesterol: total cholesterol of 235 mg/dL, LDL-C of 155 mg/dL, HDL-C of 42 mg/dL, and triglycerides of 180 mg/dL, with an untreated systolic blood pressure of 138 mmHg. He has no personal history of cardiovascular disease, but his father had a heart attack at 58, and he's a former smoker who quit five years ago. He carries some real risk factors (overweight, borderline blood pressure, family history) but doesn't fall into a clear-cut category, exactly the kind of gray-zone decision the ACC/AHA guidelines say should involve a clinician-patient discussion rather than a default prescription. Marcus is also a demanding case behaviorally: he's reluctant to be "on a pill for life," has failed at sustained diet and exercise changes twice before, but is highly motivated by his father's heart attack and genuinely worried about his own cardiovascular risk. That mix of real medical risk, and low follow-through confidence is what makes his case a useful test of the app.

2. ED² Statin Choice Application in Both Modes

a) Screenshots

User Profile:

First name: Jason
Last name: Doolittle
Email id: jdoolitt@redar.gwu.edu
[Change Password](#)

User Profile

Do you have history of cardiovascular disease: ☐ Yes ☒ No

1. Personal Attributes

Age (years):

Sex assigned at birth: ☒ Male ☐ Female

LDL-C level (mg/dL):

HDL-C level (mg/dL):

Triglycerides (mg/dL):

Total cholesterol (mg/dL):

Systolic blood pressure (mmHg):

Is blood pressure treated? ☐ Yes ☒ No

Diabetes diagnosis: ☐ Yes ☒ No

Smoking status: ☐ Current ☒ Former ☐ Never

Weight (lbs):

Height (in):

Current medications:

2. Contextual Attributes

Access to healthcare providers and medications: ☒ Good ☐ Limited ☐ Poor

Insurance coverage for medications/lifestyle programs: ☒ Yes ☐ Partial ☐ No

Medication affordability: ☒ Affordable ☐ Somewhat ☐ Not affordable

Access to diet/lifestyle counseling or programs: ☐ Yes ☒ No

Resources for lifestyle change available: ☐ Yes ☒ No

Time available for lifestyle change: ☐ High ☐ Moderate ☒ Low

Work/family responsibilities limiting treatment: ☒ Significant ☐ Manageable ☐ Minimal

3. Behavioral Attributes

Willingness to take daily medication: ☐ High ☒ Moderate ☐ Low

Confidence in maintaining lifestyle changes: ☐ High ☐ Moderate ☒ Low

Adherence to past medications: ☐ Consistent ☐ Inconsistent ☒ Never prescribed

Adherence to past lifestyle modifications: ☐ Strong ☒ Moderate ☐ Weak

Beliefs about medication: ☐ Positive ☒ Neutral ☐ Negative

Preference for non-pharmacologic treatments: ☐ Strong ☒ Moderate ☐ Weak

Risk perception (concern about future cardiovascular events): ☒ High ☐ Moderate ☐ Low

Motivation to prevent stroke/heart attack: ☒ Strong ☐ Moderate ☐ Weak

[Update Profile](#)

Subscription

Screenshot 1 — Statin Choice User Profile (Marcus)

Decision-Making Mode – Final Results (all three alternatives):

ED²®Statin Choice

Hello, Jaxon

Add problem Cancel Save

Choose a mode
Decision-making

Help me to decide

Problem: Moderate-intensity Statin therapy vs. Lifestyle therapy vs. Low-intensity statin therapy with lifestyle modification.

Alternative 1: Moderate-intensity statin therapy 71% (+64%, -27%)

Advantages
 • Relative Risk Reduction (RRR): RR = 8.5%, RRR = 30%, CL (RRR): very high
 • Substantial reduction in LDL cholesterol (30-40%), which is strongly linked to lower cardiovascular risk. CL: extremely high
 • Once-daily oral medication, requiring minimal time commitment, which fits well with limited time and significant work/family responsibilities. CL: very high
 • Well-established safety profile and strong guideline support for primary prevention in adults with elevated LDL and multiple risk factors. CL: extremely high
 • Affordable and accessible due to good insurance coverage and medication affordability. CL: extremely high

How appealing are the advantages to you? Magnitude (strong)

How likely are you to experience these advantages? Likelihood (often)

Disadvantages
 • Residual Risk (RR): RR = 5.95%, CL (RR): very high
 • Potential for side effects (e.g., muscle aches, mild liver enzyme elevations), though most are mild and reversible. CL: moderate
 • Requires daily medication adherence, which may be challenging given no prior experience with chronic medication use. CL: moderate
 • Does not address other lifestyle-related risk factors (e.g., weight, physical inactivity) unless combined with lifestyle changes. CL: high
 • May not align fully with user's moderate preference for non-pharmacologic approaches. CL: moderate

How unappealing are the disadvantages to you? Magnitude (very weak)

How likely are you to experience these disadvantages? Likelihood (seldom)

Alternative 2: Lifestyle therapy 24% (+27%, -64%)

Advantages
 • Relative Risk Reduction (RRR): RR = 8.5%, RRR = 10%, CL (RRR): moderate
 • No medication side effects or drug interactions, as this approach relies solely on lifestyle changes. CL: extremely high
 • Aligns with user's moderate preference for non-pharmacologic treatments and desire to avoid daily medication. CL: high
 • Potential for broader health benefits beyond cardiovascular risk (e.g., weight loss, improved energy, better glycemic control). CL: high
 • No cost for medications, and no need for prescription refills or medical monitoring for drug safety. CL: extremely high

How appealing are the advantages to you? Magnitude (very weak)

How likely are you to experience these advantages? Likelihood (seldom)

Disadvantages
 • Residual Risk (RR): RR = 7.65%, CL (RR): moderate
 • Lower expected risk reduction compared to statin therapy, especially given user's low confidence and limited resources for lifestyle change. CL: high
 • Requires significant time and sustained effort, which may be difficult due to work/family responsibilities and low available time. CL: very high
 • No access to structured diet/exercise counseling or programs, making effective lifestyle change less likely. CL: very high
 • That adherence to lifestyle modifications only moderate, raising concerns about long-term success. CL: high

How unappealing are the disadvantages to you? Magnitude (strong)

How likely are you to experience these disadvantages? Likelihood (often)

Alternative 3: Low-intensity statin therapy with lifestyle modification 49% (+50%, -45%)

Advantages
 • Relative Risk Reduction (RRR): RR = 8.5%, RRR = 20%, CL (RRR): high
 • Combines modest LDL reduction from statin with potential incremental benefit from lifestyle changes. CL: high
 • Lower risk of statin-related side effects compared to moderate-intensity statin therapy. CL: high
 • May better align with user's moderate preference for non-pharmacologic approaches by including lifestyle modification. CL: moderate
 • Flexible approach that allows for gradual adjustment to medication and lifestyle changes, which may improve long-term adherence. CL: moderate

How appealing are the advantages to you? Magnitude (not weak, not strong)

How likely are you to experience these advantages? Likelihood (not seldom, not often)

Disadvantages

Screenshot 2 — Statin Choice Decision-Making Mode, Alternative 1 rated (71% satisfying)

ED²®Statin Choice

Saved Problems

FAQ

User Manual

Alternative 2: Lifestyle therapy 24% (+27%, -64%)

Advantages
 • Relative Risk Reduction (RRR): RR = 8.5%, RRR = 10%, CL (RRR): moderate
 • No medication side effects or drug interactions, as this approach relies solely on lifestyle changes. CL: extremely high
 • Aligns with user's moderate preference for non-pharmacologic treatments and desire to avoid daily medication. CL: high
 • Potential for broader health benefits beyond cardiovascular risk (e.g., weight loss, improved energy, better glycemic control). CL: high
 • No cost for medications, and no need for prescription refills or medical monitoring for drug safety. CL: extremely high

How appealing are the advantages to you? Magnitude (very weak)

How likely are you to experience these advantages? Likelihood (seldom)

Disadvantages
 • Residual Risk (RR): RR = 7.65%, CL (RR): moderate
 • Lower expected risk reduction compared to statin therapy, especially given user's low confidence and limited resources for lifestyle change. CL: high
 • Requires significant time and sustained effort, which may be difficult due to work/family responsibilities and low available time. CL: very high
 • No access to structured diet/exercise counseling or programs, making effective lifestyle change less likely. CL: very high
 • That adherence to lifestyle modifications only moderate, raising concerns about long-term success. CL: high

How unappealing are the disadvantages to you? Magnitude (strong)

How likely are you to experience these disadvantages? Likelihood (often)

Alternative 3: Low-intensity statin therapy with lifestyle modification 49% (+50%, -45%)

Advantages
 • Relative Risk Reduction (RRR): RR = 8.5%, RRR = 20%, CL (RRR): high
 • Combines modest LDL reduction from statin with potential incremental benefit from lifestyle changes. CL: high
 • Lower risk of statin-related side effects compared to moderate-intensity statin therapy. CL: high
 • May better align with user's moderate preference for non-pharmacologic approaches by including lifestyle modification. CL: moderate
 • Flexible approach that allows for gradual adjustment to medication and lifestyle changes, which may improve long-term adherence. CL: moderate

How appealing are the advantages to you? Magnitude (not weak, not strong)

How likely are you to experience these advantages? Likelihood (not seldom, not often)

Disadvantages

Screenshot 2b — Statin Choice Decision-Making Mode, Alternatives 2 and 3 rated (24% and 49%)

Reactive Problem-Solving Mode – Final Results (all three alternatives):

ED2® Statin Choice

Hello, Jaxon

Add problem

Choose a mode
Reactive problem-solving

Help me to decide

Problem: Moderate-intensity Statin therapy vs. Lifestyle therapy vs. Low-intensity statin therapy with lifestyle modification.

Present risk event: high cholesterol

Risk: risk of heart attack and stroke

Activity goal: lowering LDL-C by 30%.

Problem goal: sufficiently reducing the risk of heart attack and stroke

Alternative 1: Moderate-intensity statin therapy 64% (+64%, -36%)

Hypothesis 1: helps in sufficiently reducing the risk of heart attack and stroke 64% (+64%, -36%)

Hypothesis 2: doesn't help in sufficiently reducing the risk of heart attack and stroke 64% (+64%, -36%)

Alternative 2: Lifestyle therapy 30% (+27%, -48%)

Hypothesis 1: helps in sufficiently reducing the risk of heart attack and stroke 22% (+36%, -77%)

Hypothesis 2: doesn't help in sufficiently reducing the risk of heart attack and stroke 39% (+18%, -18%)

Alternative 3: Low-intensity statin therapy with lifestyle modification 25% (+32%, -46%)

Hypothesis 1: helps in sufficiently reducing the risk of heart attack and stroke 22% (+36%, -77%)

Hypothesis 2: doesn't help in sufficiently reducing the risk of heart attack and stroke 27% (+27%, -55%)

Notes: Deciding whether to take statins or implement lifestyle therapy

Auto Generate Pros & Cons

Screenshot 3 — Statin Choice Reactive Problem-Solving Mode, final results (64%, 30%, 25%)

b) Evidence of Self-Efficacy

Coping self-efficacy. While rating the likelihood of the Hypothesis 1 disadvantages (difficulty in lowering LDL-C by 30%) for Alternative 1, moderate-intensity statin therapy, the AI surfaced material, physical, psychological, social, and spiritual challenges, things like unfamiliarity with chronic medication, possible mild side effects, and work/family responsibilities interfering with a daily routine. Even though these are real concerns for Marcus, I rated the magnitude as weak and the likelihood as seldom, because the app's own suggested solutions (setting reminders, integrating the pill into an existing routine like breakfast, working with his provider on any side effects) are realistic and low effort. That's coping self-efficacy in action: my confidence that Marcus can manage the difficulty directly lowered how likely I judged it to actually derail him.

ED2® Statin Choice

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Present risk event: high cholesterol

Risk: risk of heart attack and stroke

Activity goal: lowering LDL-C by 30%

Problem goal: sufficiently reducing the risk of heart attack and stroke

Alternative 1: Moderate-intensity statin therapy 64% (+64%, -36%)

Hypothesis 1: helps in sufficiently reducing the risk of heart attack and stroke 64% (+64%, -36%)

Advantages

Relative Risk Reduction (RRR): RR = 8.5%, RRR = 30%, CL(RRR): very high

How are the advantages appealing to you from the perspective of sufficiently reducing the risk of heart attack and stroke?

Magnitude (strong)

How likely are you to experience these advantages?

Likelihood (often)

Disadvantages

Difficulty in lowering LDL-C by 30%

Medication cost is not a barrier due to insurance and affordability, but requires ongoing prescription and pharmacy visits. CL: extremely low. Solution: Utilize insurance and pharmacy delivery services to minimize inconvenience.

Possible mild side effects (muscle aches, liver enzyme changes), but most are manageable. CL: moderate. Solution: Regular monitoring and prompt reporting of symptoms to healthcare provider.

Moderate willingness to take daily medication, but no prior adherence to chronic meds may cause anxiety or uncertainty. CL: high. Solution: Education and support from provider to build confidence and routine.

Work/family responsibilities may interfere with medication routine. CL: very high. Solution: Set reminders and integrate medication into daily schedule (e.g., with breakfast).

No specific spiritual barriers identified. CL: extremely low. Solution: If concerns arise, discuss with provider to align treatment with personal values.

How are the disadvantages unappealing to you from the perspective of sufficiently reducing the risk of heart attack and stroke?

Magnitude (weak)

How likely are you to experience these disadvantages?

Likelihood (seldom)

Hypothesis 2: doesn't help in sufficiently reducing the risk of heart attack and stroke 64% (+64%, -36%)

Alternative 2: Lifestyle therapy 30% (+27%, -48%)

Hypothesis 1: helps in sufficiently reducing the risk of heart attack and stroke 22% (+36%, -77%)

Hypothesis 2: doesn't help in sufficiently reducing the risk of heart attack and stroke 39% (+18%, -18%)

Alternative 3: Low-intensity statin therapy with lifestyle modification 25% (+32%, -66%)

Screenshot 4 — Coping Self-Efficacy: rating likelihood of difficulty (weak / seldom)

Self-regulatory efficacy. While rating the likelihood of the Hypothesis 2 advantages (the significance of successfully lowering LDL-C by 30%) for the same alternative, the AI generated strong physical and psychological benefits, substantially reduced cardiovascular risk and reduced anxiety about future events. I rated the magnitude as strong and the likelihood as often, reflecting my confidence that Marcus, given his high stated concern about his father's heart attack and strong motivation to prevent his own, would actually follow through consistently enough to realize those benefits. That's self-regulatory efficacy: belief in Marcus's ability to sustain the goal-directed effort raised how likely I judged the benefits to occur.

ED2® Statin Choice

Saved Problems
FAQ
User Manual

Hypothesis 2: doesn't help in sufficiently reducing the risk of heart attack and stroke 64% (+64%, -36%)

Advantages
Significance of lowering LDL-C by 30%.

- Statistical:** Lowering LDL-C by 30% reduces medication costs for future cardiovascular events and hospitalizations. Strength: strong. EL: high. Strategy: Continue statin therapy to prevent costly complications.
- Physiological:** Significant reduction in physical risk of heart attack and stroke, improving longevity and quality of life. Strength: very strong. EL: extremely high. Strategy: Maintain statin adherence and monitor health.
- Psychological:** Reduces anxiety about future cardiovascular events, supporting mental well-being. Strength: strong. EL: high. Strategy: Reinforce risk reduction benefits during follow-up.
- Social:** Improved health may enhance ability to fulfill work/family responsibilities. Strength: moderate. EL: moderate. Strategy: Share progress with family/support network.
- Personal:** May align with personal values of prevention and responsibility for health. Strength: moderate. EL: moderate. Strategy: Reflect on health goals and discuss with provider if needed.

How are the advantages appealing to you from the perspective of sufficiently reducing the risk of heart attack and stroke?
Magnitude (strong)

How likely are you to experience these advantages?
Likelihood (often)

Disadvantages
Residual Risk (RR): RR = 5.95%, OL(RR): very high

How are the disadvantages unappealing to you from the perspective of sufficiently reducing the risk of heart attack and stroke?
Magnitude (weak)

How likely are you to experience these disadvantages?
Likelihood (seldom)

Alternative 2: Lifestyle therapy 30% (+27%, -48%)

Hypothesis 1: helps in sufficiently reducing the risk of heart attack and stroke 22% (+36%, -77%)

Hypothesis 2: doesn't help in sufficiently reducing the risk of heart attack and stroke 39% (+18%, -18%)

Alternative 3: Low-intensity statin therapy with lifestyle modification 25% (+32%, -66%)

Hypothesis 1: helps in sufficiently reducing the risk of heart attack and stroke 22% (+36%, -77%)

Screenshot 5 — Self-Regulatory Efficacy: rating likelihood of significance (strong / often)

Table 1 summarizes the final satisficing levels across both modes.

Alternative	Decision-Making Mode	Reactive Problem-Solving Mode
Alt 1: Moderate-intensity statin therapy	71% (+64%, -27%)	64% (+64%, -36%)
Alt 2: Lifestyle therapy	24% (+27%, -64%)	30% (+27%, -48%)
Alt 3: Low-intensity statin + lifestyle modification	49% (+50%, -45%)	25% (+32%, -66%)

3. Discussion Questions

a) Consistency among BR, RRR, RR, and Confidence Levels; any inconsistencies or hallucinations?

The Baseline Risk stayed locked at 8.5% across all three alternatives and both modes, which is what you'd expect since BR reflects Marcus's underlying profile, not the alternative being evaluated. The Residual Risk math checked out arithmetically every time: $RR = BR \times (1 - RRR)$ gave 5.95% for Alternative 1 ($8.5\% \times 0.70$), 7.65% for Alternative 2 ($8.5\% \times 0.90$), and 6.8% for Alternative 3 ($8.5\% \times 0.80$), all matching the numbers the app displayed. The Confidence Levels also tracked sensibly with the strength of the underlying evidence, CL was "extremely high" on well-established claims like statin safety and guideline support and dropped to "moderate" on softer claims about lifestyle-adherence outcomes.

Where I did notice a real inconsistency was in how much the final satisficing percentages shifted between modes for the same alternative, especially Alternative 3, which dropped from 49% in decision-making mode to 25% in reactive problem-solving mode, a 24-point swing. Both modes still agreed on the overall winner (Alternative 1), and the margin over the runner-up actually got larger in problem-solving mode (34 points vs. 22), so the final recommendation was consistent even though the underlying numbers for the losing alternatives moved around more than I expected. I didn't see anything I'd call an outright hallucination, no numbers that contradicted each other within the same screen, but the app never explains why the same alternative scores so differently once you move from the aggregate advantage/disadvantage view to the hypothesis-based breakdown, which makes it harder to trust that the two modes are really measuring the same thing.

b) Most prominent categories of difficulty and significance; how helpful were the AI-generated solutions?

Psychological and social difficulty were the most prominent categories for Marcus, anxiety about starting a medication he's never taken before, and work/family responsibilities interfering with building a new daily routine. That tracks directly with his profile: low confidence in maintaining lifestyle changes and significant work/family load. On the significance side, physical and psychological significance dominated, meaningfully reduced cardiovascular risk and reduced anxiety about a future heart attack, both of which map straight onto his high concern about repeating his father's history.

The AI-generated solutions were genuinely specific rather than generic. "Set reminders and integrate medication into daily schedule (e.g., with breakfast)" for the social/routine difficulty and "regular monitoring and prompt reporting of symptoms to healthcare provider" for physical side effects both read like something a clinician would actually say, not boilerplate advice. That specificity is what made it reasonable to rate several of the difficulties as low likelihood despite being real concerns on paper.

c) Intuitive choice vs. AI-favored alternative vs. final decision

Going in, my intuitive read of Marcus's case was that he'd lean toward lifestyle therapy, his profile is built around a stated preference for non-pharmacologic treatment and real reluctance about lifelong medication. The app's AI-favored alternative, once actual numbers were on the table, was clearly Alternative 1 (moderate-intensity statin) in both modes, by a wide and consistent margin. My final decision matched the app's recommendation, not my initial intuition.

The reason the two diverged is worth calling out: lifestyle therapy's RRR (10%) was roughly a third of the statin's (30%), and that gap only got more important once I weighed it against Marcus's own documented track record, he'd already failed at sustained lifestyle change twice. Once the evidence-weighted risk reduction and the behavioral profile both pointed in the same direction, my preference-driven intuition (avoid the pill) didn't hold up. That's a fairly direct illustration of what Lehto and Nanda and the DI course materials describe: descriptively, people default to intuitive, preference-driven choices, but a structured decision augmentation process forces a more normative, evidence-weighted evaluation that can override the initial gut call.

d) Extent to which generated outcomes reflected the User Profile

Very closely. The AI explicitly wove specific profile fields into its generated text rather than producing generic statin pros and cons, phrases like "requiring minimal time commitment, which fits well with limited time and significant work/family responsibilities," "affordable and accessible due to good insurance coverage and medication affordability," and "may not align fully with user's moderate preference for non-pharmacologic approaches" all tie directly back to specific fields I entered (time availability, insurance coverage, stated treatment preference). That level of specificity is what made the rating process feel meaningful rather than arbitrary.

e) Recommended improvements

The rating sliders apply a single Magnitude/Likelihood pair to an entire block of five advantages or five disadvantages at once, rather than letting the user rate each bullet individually. That's faster to use,

but it forces very different concerns, for example, “no cost for medication” and “no side effects” under Lifestyle therapy's advantages, into a single averaged rating, which loses precision exactly where the app is trying to be personalized. Letting users rate difficulty and significance by individual category, the way the reactive mode's hypothesis structure already implies, would make the underlying satisficing math more defensible.

I'd also recommend the app surface an explanation whenever the same alternative's satisficing level shifts significantly between decision-making and problem-solving mode (as Alternative 3 did here, 49% to 25%), so users understand whether that's a meaningful change in the underlying risk model or simply an artifact of the more granular rating structure in reactive mode.

Task 2: ED² Insurance Choice – Reputation Insurance Decision-Making and Proactive Problem-Solving

1. Brief Description of Case 2

For Task 2 I was assigned Reputation insurance. The case is built around a founder profile modeled on my own background running an early-stage AI startup: a public-facing role with press interviews and an active social media presence, an AI product that makes automated decisions for customers, and no dedicated PR or legal staff to absorb a crisis if one hits. The estimated exposure runs \$300,000–\$500,000 in potential lost revenue, partnerships, and fundraising ability if a reputational crisis occurs disproportionate to the company's size given how quickly a viral complaint or an AI mistake can spread. Financial capacity is moderate: the startup has limited cash reserves and would strain its budget to self-fund a crisis response. Risk tolerance is moderate rather than high, despite general comfort with financial risk from a trading and startup background, because reputational damage is slower to reverse and compounds across investors, customers, and press simultaneously. The decision preference is a balanced approach conscious but not cost-first, and the engagement style is an active researcher who occasionally consults an advisor.

2. ED² Insurance Choice Application in Both Modes

a) Screenshots

User Profile:

Personal Info

First name:
Last name:
[Change Password](#)

Subscription

Plan name:
Expires on: Jul 23, 2025
Renews on: Jul 23, 2025
[Change Plan](#)

Insurance Profile

Complete your profile so we can tailor your insurance decision analysis.

1. Asset-at-Risk
Select asset type
Reputation

2. Estimated Asset-at-Risk Value / Exposure
☐ Low
☐ Moderate
☒ High

3. Financial Capacity
☐ Low
☒ Moderate
☐ High

4. Geographic / Environmental / Situational Risk
☐ Low
☐ Moderate
☒ High
(describe (optional))
Public-facing AI startup founder; product makes automated decisions for customers; no in-ho

5. Risk Tolerance
☐ Low
☒ Moderate
☐ High

6. Decision Preferences
☐ Cost-focused
☐ Comprehensive protection
☒ Balanced approach

7. Engagement Style
☒ Active researcher
☐ Advisor-dependent
☐ Minimal involvement

[Save Profile](#) [Next](#)

Screenshot 6 — Insurance Choice User Profile (Reputation)

Decision-Making Mode – Final Results (all three coverage alternatives):

jsoulin@radar.gsw.edu

Add problem Cancel Save

Choose a mode
Decision Making

Help me to decide Pros & Cons Generated

Problem: Reputation: Basic vs Standard vs Comprehensive

Risk: risk of financial loss

Basic: Basic Coverage (Crisis communications; media monitoring; \$250k-\$500k limits)	37% (+50%, -64%)	✓ ✎ 🗑
Standard: Expanded Coverage (Crisis response + lost income + liability defense; \$1M+ limits)	58% (+64%, -50%)	✓ ✎ 🗑
Comprehensive: Comprehensive Coverage (Global PR + income protection + cyber-linked harm; \$5M+ limits)	51% (+55%, -50%)	✓ ✎ 🗑

Screenshot 7 — Insurance Choice Decision-Making Mode, final results (37%, 58%, 51%)

Proactive Problem-Solving Mode – Final Results (all three coverage alternatives):

jsoulin@radar.gsw.edu

Add problem Cancel Save

Choose a mode
Problem Solving

Help me to decide Pros & Cons Generated

Problem: Reputation: Basic vs Standard vs Comprehensive

Future risk event: Defamation, misinformation, online attacks, negative media, social-media crises, or scandals causing loss of trust, revenue, or brand value

Risk: risk of financial loss

Activity goal: receiving financial support after a covered loss

Problem goal: sufficiently reducing the risk of financial loss

Basic: Basic Coverage (Crisis communications; media monitoring; \$250k-\$500k limits)	31% (+39%, -57%)	
Hypothesis 1: 1: Defamation, misinformation, online attacks, negative media, social-media crises, or scandals causing loss of trust, revenue, or brand value happens	37% (+50%, -64%)	
Hypothesis 2: 2: Defamation, misinformation, online attacks, negative media, social-media crises, or scandals causing loss of trust, revenue, or brand value doesn't happen	29% (+27%, -50%)	
Standard: Expanded Coverage (Crisis response + lost income + liability defense; \$1M+ limits)	54% (+64%, -55%)	
Hypothesis 1: 1: Defamation, misinformation, online attacks, negative media, social-media crises, or scandals causing loss of trust, revenue, or brand value happens	59% (+64%, -45%)	
Hypothesis 2: 2: Defamation, misinformation, online attacks, negative media, social-media crises, or scandals causing loss of trust, revenue, or brand value doesn't happen	51% (+64%, -64%)	
Comprehensive: Comprehensive Coverage (Global PR + income protection + cyber-linked harm; \$5M+ limits)	58% (+64%, -48%)	
Hypothesis 1: 1: Defamation, misinformation, online attacks, negative media, social-media crises, or scandals causing loss of trust, revenue, or brand value happens	59% (+64%, -45%)	
Hypothesis 2: 2: Defamation, misinformation, online attacks, negative media, social-media crises, or scandals causing loss of trust, revenue, or brand value doesn't happen	58% (+64%, -50%)	

Screenshot 8 — Insurance Choice Proactive Problem-Solving Mode, final results (31%, 54%, 58%)

b) Evidence of Self-Efficacy

Coping self-efficacy. While evaluating the Hypothesis 1 disadvantages (difficulty in receiving financial support after a covered loss) for Basic coverage, the AI flagged material, physical, psychological, social, and spiritual challenges, limited financial coverage not fully addressing high-value reputation risk, stress from managing a crisis with limited support, anxiety over insufficient coverage for major incidents, and strain on professional relationships. Unlike the Statin Choice example, here I rated the magnitude as strong and the likelihood as often, because these difficulties are structural rather than something confidence alone can fix: Basic coverage genuinely does not address the cyber-linked harm and global PR exposure central to an AI startup founder's risk profile, and none of the app's suggested solutions (supplementing with savings, building an advisor network) close that structural gap. That's coping self-efficacy working in the other direction, low confidence that the difficulty can be managed pushed the likelihood rating up, which is part of why Basic coverage's overall satisficing level (31–37%) landed well below the 51% threshold.

Future risk event: Defamation, misinformation, online attacks, negative media, social-media crises, or scandals causing loss of trust, revenue, or brand value

Risk: risk of financial loss

Activity goal: receiving financial support after a covered loss

Problem goal: sufficiently reducing the risk of financial loss

Basic: Basic Coverage (Crisis communications; media monitoring; \$250k–\$500k limits) 31% (+39%, -57%)

Hypothesis 1: 1: Defamation, misinformation, online attacks, negative media, social-media crises, or scandals causing loss of trust, revenue, or brand value happens 37% (+50%, -64%)

Advantages
Describe the advantages:
RR = 8%, RRR = 30%, CL(RRR): moderate

How are the advantages appealing to you from the perspective of sufficiently reducing the risk of financial loss?
Magnitude (not weak, not strong)

How likely are you to experience these advantages?
Likelihood (not seldom, not often)

Difficulty in receiving financial support after a covered loss
Describe the disadvantages:
RR = 5.6%, CL(RR): moderate

Material:
Limited financial coverage may not fully address high-value reputation risks. [CL: moderate] – Solution: Supplement with additional savings or external PR resources.

Physical:
Stress-related health impacts from managing crises with limited support. [CL: moderate] – Solution: Adopt stress management techniques and seek professional health support.

Psychological:
Increased anxiety due to insufficient coverage for major incidents. [CL: moderate] – Solution: Engage in proactive mental health counseling and crisis preparation.

Social:
Potential strain on professional relationships due to limited crisis response. [CL: moderate] – Solution: Build a network of trusted advisors and collaborators for support.

Spiritual:
Feelings of inadequacy in protecting personal and professional values. [CL: moderate] – Solution: Seek guidance from mentors or spiritual advisors to align actions with values.

How are the disadvantages unappealing to you from the perspective of sufficiently reducing the risk of financial loss?
Magnitude (strong)

How likely are you to experience these disadvantages?
Likelihood (often)

Hypothesis 2: 2: Defamation, misinformation, online attacks, negative media, social-media crises, or scandals causing loss of trust, revenue, or brand value doesn't happen 29% (+27%, -50%)

Screenshot 10 — Coping Self-Efficacy: rating likelihood of difficulty (strong / often), Basic coverage

Self-regulatory efficacy. While evaluating the Hypothesis 2 advantages (the significance of receiving financial support while sufficiently reducing risk) for Comprehensive coverage, the AI generated its strongest-worded significance language across every category “maximum financial security,” “highest level of peace of mind,” “enhanced reputation as a highly prepared and responsible professional.” Rating the likelihood of those benefits as often reflects confidence that the founder would actually follow through on maintaining the coverage and leveraging it, communicating preparedness to stakeholders, reflecting on long-term goals, rather than letting the policy sit unused. That's the self-regulatory efficacy component: belief in sustained follow-through raising the likelihood rating for the benefit.

Full alignment with values of comprehensive protection and responsibility [CL: moderate] – Solution: Reinforce alignment through regular reflection on personal and professional goals.

How are the disadvantages unappealing to you from the perspective of sufficiently reducing the risk of financial loss?

Magnitude (not weak, not strong)

How likely are you to experience these disadvantages?

Likelihood (seldom)

Hypothesis 2: 2: Defamation, misinformation, online attacks, negative media, social-media crises, or scandals causing loss of trust, revenue, or brand value doesn't happen 58% (+64%, -50%)

Significance of receiving financial support after a covered loss

Describe the advantages

BR = 8%, RRR = 90%, CL(RRR): moderate

Material

Maximum financial security even without a crisis. [Strength: very strong, EL: moderate] – Strategy: Maintain coverage to ensure long-term protection.

Physical

Minimal physical stress due to comprehensive preparedness. [Strength: very strong, EL: moderate] – Strategy: Focus on maintaining a healthy lifestyle.

Psychological

Highest level of peace of mind from extensive protection. [Strength: very strong, EL: moderate] – Strategy: Leverage coverage to focus on growth and innovation.

Reputational

Enhanced reputation as a highly prepared and responsible professional. [Strength: very strong, EL: moderate] – Strategy: Communicate preparedness to stakeholders to build trust.

Personal

Complete alignment with values of security and foresight. [Strength: very strong, EL: moderate] – Strategy: Reaffirm commitment to long-term goals and values.

How are the advantages appealing to you from the perspective of sufficiently reducing the risk of financial loss?

Magnitude (strong)

How likely are you to experience these advantages?

Likelihood (often)

Disadvantages

Describe the disadvantages

RR = 0.8%, CL(RR): moderate

How are the disadvantages unappealing to you from the perspective of sufficiently reducing the risk of financial loss?

Magnitude (not weak, not strong)

How likely are you to experience these disadvantages?

Likelihood (not seldom, not often)

Screenshot 9 — Self-Regulatory Efficacy: rating likelihood of significance (strong / often)

Table 2 summarizes the final satisficing levels across both modes.

Coverage Alternative	Decision-Making Mode	Proactive Problem-Solving Mode
Basic (Crisis comms + media monitoring; \$250k–\$500k limits)	37% (+50%, –64%)	31% (+39%, –57%)
Standard (Expanded coverage; \$1M+ limits)	58% (+64%, –50%)	54% (+64%, –55%)
Comprehensive (Global PR + cyber-linked harm; \$5M+ limits)	51% (+55%, –50%)	58% (+64%, –48%)

3. Discussion Questions

a) Consistency among BR, RRR, RR, and Confidence Levels; any inconsistencies or hallucinations?

Baseline Risk held steady at 8% across all three coverage tiers and both modes. The Residual Risk math checked out arithmetically in every case: $RR = BR \times (1 - RRR)$ gave 5.6% for Basic ($8\% \times 0.70$), 2.4% for Standard ($8\% \times 0.30$), and 0.8% for Comprehensive ($8\% \times 0.10$), matching the app's displayed figures exactly.

The clearest inconsistency I found was in the Difficulty sections (Hypothesis 1 disadvantages) for Standard and Comprehensive coverage. Several of the auto-generated entries under the Physical, Psychological, Social, and Spiritual categories actually described positive outcomes rather than genuine difficulties, for example, “Enhanced ability to maintain professional relationships during crises,” “Full alignment with values of comprehensive protection and responsibility,” and “Significant peace of mind from extensive coverage,” all listed under Difficulty. Those read like benefits that got mislabeled, not real challenges. It looks like the AI ran short of genuine difficulty content for the higher coverage tiers (where there genuinely is less to complain about) and defaulted to restating advantages under the wrong heading rather than acknowledging that a tier has minimal difficulty. That's a real content-generation inconsistency worth flagging, even though the underlying facts weren't factually wrong.

b) Most prominent categories of difficulty and significance for Reputation insurance; how helpful were the AI-generated solutions?

Material difficulty was the dominant, most legitimate concern across all three tiers. Basic's limited coverage not addressing the real cyber-linked harm and global PR exposure of an AI startup, and Standard/Comprehensive's premium cost straining the profile's stated moderate financial capacity. That tension between what full protection costs and what the profile says is affordable is the central conflict running through this case. Psychological significance, peace of mind and confidence, was the dominant benefit category once meaningful coverage was in place, especially for Comprehensive.

The solutions were reasonable but more generic than I'd like for a case this specific. “Budget for premiums and explore flexible payment options” and “evaluate cost-benefit trade-offs and explore financing options” are sound general advice, but they don't get specific to an early-stage startup's actual financing tools, bundling reputation coverage into a broader D&O or tech E&O policy, or negotiating a premium schedule tied to funding milestones, would have been more useful and more personalized than what the app generated.

c) Intuitive choice vs. AI-favored alternative vs. final decision

My intuitive pick going in was comprehensive, given the startup's real exposure to cyber-linked harm and global PR risk, the highest tier felt like the obviously safe answer. Decision-making mode actually favored Standard (58%) over Comprehensive (51%), a 7-point gap. Problem-solving mode narrowed that gap further, with Comprehensive edging ahead at 58% versus Standard's 54%, a 4-point margin that falls inside the 5% proximity threshold AHFE papers describe for this framework, even though the interface itself never explicitly flagged a proximity alert.

My final decision landed on Standard, not my original intuitive pick of Comprehensive. The deciding factor was the material-difficulty theme both modes kept surfacing: Comprehensive's premium genuinely strains the profile's stated moderate financial capacity, while Standard already closes most of the risk gap (RRR 70% vs. 90%, RR 2.4% vs. 0.8%) and comfortably clears the 51% satisficing threshold on its own. This is a good real-world example of the tension the Insurance Choice paper describes between cognitive risk reduction and motivational/financial constraints, the numerically “safest” option isn't automatically the satisficing one once actual financial capacity gets weighed into the picture.

d) Extent to which generated outcomes reflected the User Profile

Strongly. The generated content repeatedly referenced my exact profile fields by name, “aligning with the user's moderate risk tolerance and balanced decision preferences,” “supporting the user's active engagement style,” “addressing critical risks for the user's AI startup environment,” and “may exceed the user's moderate financial capacity.” The Geographic/Situational Risk description I typed into the profile, public-facing AI founder, no in-house PR or legal team, showed up nearly verbatim in some of the generated advantage and disadvantage text for Comprehensive coverage. That's a tight enough match to confirm the profile fields are directly driving the underlying prompt, not just cosmetically referenced somewhere in the output.

e) Recommended improvements

First, fix the Difficulty/Significance mislabeling identified in 3a the app should either generate genuinely negative content for every coverage tier's Difficulty section or explicitly state when a tier has minimal difficulty, rather than restating advantages under the wrong heading. Second, the proximity between Standard and Comprehensive (4–7 points across both modes) never triggered a visible proximity alert, even though it fell well within the threshold the app's own underlying framework describes. Surfacing that alert explicitly and prompting the user toward the “refine your decision” step described in the manual adding a new objective to sharpen the Problem Goal would make the tool's stated methodology more visible and would have helped resolve the Standard-vs-Comprehensive tension more systematically instead of leaving it to my own judgment call.